

Percent Change T-test

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2026-05-18

Set up

```
library(tidyverse)
library(janitor)
library(ggrepel)
library(cowplot)
library(here)
library(ggtext)
library(flextable)
```

```
# read in data
veg <- read_csv(here("data", "raw", "veg.csv"))

metadata <- read_csv(here("data", "raw", "vp_veg_metadata.csv"))
```

Creating gl_explicit0

```
poaceae_species <- c(
  "Bromus_hordeaceus",
  "Festuca_perennis",
  "Stipa_pulchra",
  "Polypogon_monspeliensis",
  "Brachypodium_distachyon",
  "Festuca_myuros",
  "Avena_barbata",
  "Hordeum_marinum",
```

```

"Avena_fatua",
"Parapholis_incurva",
"Bromus_diandrus",
"Distichlis_spicata",
"Bromus_catharticus",
"Hordeum_murinum",
"Hordeum_brachyantherum_brachy.",
"Polypogon_viridis",
"Cortaderia_selloana",
"Festuca_bromoides",
"Cynodon_dactylon",
"Poa_annua",
"Festuca_microstachys",
"Crypsis_schoenoides",
"Polypogon_interruptus",
"Hordeum_brachyantherum",
"Bromus_madritensis",
"Elymus_triticoides"
)

native_forbs <- c(
  "Plantago_erecta",
  "Sisyrinchium_bellum",
  "Sidalcea_malviflora",
  "Eryngium_vaseyi",
  "Acmispon_maritimus",
  "Acmispon_wrangelianus",
  "Madia_sativa",
  "Psilocarphus_tennulus",
  "Psilocarphus_brevissimus",
  "Spergularia_macrotheca",
  "Centromadia_parryi_australis",
  "Lepidium_nitidum",
  "Stebbinsoseris_heterocarpa",
  "Laennecia_coulteri"
)

exclude_other <- c(
  "Bare Ground",
  "Natural Thatch",
  "unlisted",
  "moss",

```

```

    "boulder_rock",
    "biocrust",
    "Baccharis_pilularis",
    "Juncus_bufonius",
    "Juncus_phaeocephalus",
    "Eleocharis_macrostachya",
    "Eleocharis_acicularis",
    "Eleocharis_palustris"
  )

```

```

species_lookup <- veg |>
  # keep only GL transects at ncos
  filter(str_detect(transect_name, "GL"), site == "ncos") |>
  # remove unlisted since it maps to multiple categories
  filter(psoc != "unlisted") |>
  # return one unique row per species-category combination
  distinct(psoc, cover_category) |>
  # sort alphabetically by species for readability
  arrange(psoc)

```

```

# create a new object from veg
gl_explicit0 <- veg |>
  # filter all occurrences where transect_name contains VP
  filter(str_detect(transect_name, "GL"),
    site == "ncos") |>
  # not including cover_category because we will join it back after complete()
  select(-cover_category) |>
  # complete all possible combinations of year, pool, psoc and distance
  complete(year, transect_name, psoc, transect_distance,
    # fill values of percent cover with 0
    fill = list(percent_cover = 0)) |>
  # group by year, pool, and psoc
  group_by(year, transect_name, psoc) |>
  # calculate total percent cover across all quads
  summarize(sum_pc = sum(percent_cover, na.rm = TRUE)) |>
  # ungroup the data frame
  ungroup() |>
  # create a new column called year_pool from year and transect name
  unite("year_pool", year, transect_name, remove = TRUE) |>
  # join with metadata data frame
  left_join(metadata, by = "year_pool") |>
  # calculate mean percent cover from total percent cover/number of quadrats

```

```
mutate(mean_pc = sum_pc/num_quad) |>
# re-attach cover_category from lookup using psoc as key
left_join(species_lookup, by = "psoc") |>
# excluding "GL-11" because it was unburned
filter(transect_name != "GL-11")
```

Creating percent change data frame

```
## Change in percentage points of focal species percent cover
## 2023 = pre-burn
## 2024 = post-burn

focal_species <- tibble(
  psoc = c(
    "Stipa_pulchra",
    "Medicago_polymorpha",
    "Melilotus_indicus",
    "Festuca_perennis"
  ),
  species_label = c(
    "S. pulchra",
    "M. polymorpha",
    "M. indicus",
    "F. perennis"
  ),
  status = c(
    "Native",
    "Invasive",
    "Invasive",
    "Invasive"
  )
)

focal_cover <- gl_explicit0 |>
  filter(
    year %in% c(2023, 2024),
    psoc %in% focal_species$psoc
  ) |>
  select(year, transect_name, psoc, mean_pc) |>
  left_join(focal_species, by = "psoc")
```

```
focal_change_transect_pp <- focal_cover |>
  pivot_wider(
    names_from = year,
    values_from = mean_pc,
    names_prefix = "year_"
  ) |>
  mutate(
    percent_point_change = year_2024 - year_2023
  ) |>
  filter(!is.na(percent_point_change))

focal_change_summary_pp <- focal_change_transect_pp |>
  group_by(psoc, species_label, status) |>
  summarize(
    mean_percent_point_change = mean(percent_point_change, na.rm = TRUE),
    n = n(),
    .groups = "drop"
  ) |>
  mutate(
    species_label = factor(
      species_label,
      levels = c(
        "S. pulchra",
        "M. polymorpha",
        "M. indicus",
        "F. perennis"
      )
    ),
    status = factor(status, levels = c("Native", "Invasive"))
  )
```

1. Mean Percent Cover T-test

Table 1: Paired t-test results comparing mean percent cover of four focal species before (2023) and after (2024) the prescribed burn along Mesa (GL) transects at NCOS (n = 7 transects). Significant p-values ($p < 0.05$) are bolded. Tests were conducted without correction for multiple comparisons.

Species	t-statistic	df	p-value
<i>F. perennis</i>	1.013	6	0.350
<i>M. polymorpha</i>	0.381	6	0.716
<i>M. indicus</i>	-3.508	6	0.013
<i>S. pulchra</i>	4.658	6	0.003